tăng công suất 40GW - trang 1 (46) và biểu đồ 1 ở trang 4 (49)

sự sụt giảm chi phí công nghệ và silicon: trang 48 (3) và trang 49 (4)

Châu Âu chiếm ~85% công suất, Đức dẫn đầu: trang 1

Điện mặt trời sản xuất khoảng 50 TWh: trang 1

Solar (PV) hit the heights in 2010 despite market concerns

FORECASTS FOR PV growth rates early in 2010 were fuelled by the continuous increases that had been witnessed over the previous few years. And the final installed capacity figures at the end of last year didn't disappoint; global **cumulative installed capacity** reached almost 40 GW - a stunning gain of 72% from the previous year. **Newly-installed capacity** increased by 135% on 2009 (16.6 GW).

Europe maintained its leading market position for 2010 with around 85% of global capacity installed in the region (though this meant mainly Germany, Spain and Italy). Around 9% was installed in Asian countries, while North America's cumulative installations amounted to around 8% (the majority in one State, California).

Thirty (30) GW of the world's total solar PV power installed by the end of 2010 was located in only five countries (*see figure 1*). The estimated total electricity generated by solar PV plants reached 50 TWh (*see table 3*).

One major global trend in PV during 2010 was the increase in size and scale of grid-connected ground mounted PV power plants. The largest PV plant connected to the grid at the end of 2010 was the 97 MW *Sarnia* PV power plant in Canada (whose commercial start up was phased in during 2009 and 2010); followed by the 84 MW *Montalto di Castro* plant in Italy (also constructed in 2009 and 2010). Third was the 80MW *Finsterwalde* solar park in Germany, which uses **Q-Cells** modules.

This tendency towards larger size PV plants seems set to continue, with the development of several projects in the 100 MW range (and higher). One example of this is Masdar's *Nour One* project in Abu Dhabi, which is expected to be constructed in 2012/2013.

Few markets remain dominant

Germany remained the most important solar PV market in 2010, with 45% of the newly-installed capacity worldwide. However, in light of the 13% reduction in the FiTs (feed-in-tarifs) of January 2011, and another possible reduction planned for later this year, it is questionable whether Germany will be able to continue to cement its position as the leading market going forward (*Ed – this seems to be backed up by disappointing installation figures for early 2011, though things may be picking up. See Paula Mints' 'Comment' on pages 36-37*).

As we go to press it appears that the German government may be about to backtrack on the additional FiT cut planned for 1 July, because of disappointing installation figures in the first half of the year (*source – Reuters*). Between March and May 2011, around 700MW of new solar capacity was installed, and Government officials are reportedly wary of putting even more pressure on the PV industry. System installations for the March to May period will set the projections for the whole year, and the industry's failure to meet or exceed 3.5 GW means that the flexible part of the annual gradual decrease will not be activated on 1 July and 1 September. Consequently the FiT will not be subjected to the anticipated additional 15% cut this year.

With this uncertainty in mind, forecasting new capacity for 2011 is especially difficult. The spectre of potential FiT reductions will have an effect on supply/demand - and therefore module prices - as the effect of changes (not to mention the threat of "proposed" changes) to the scheme are gradually reflected in buying patterns.

While in earlier years a large proportion of Germany's installed capacity was rooftop based, larger, ground-mounted plants played a more important role in 2010. More than 40% of all German installations were 100 kWp or larger (although among these, there is still a substantial amount of roof-top projects, since these are getting larger as well).

The largest ground-mounted projects installed in 2011 in Germany were:

This tendency towards larger size PV plants seems set to continue, with the development of several projects in the 100 MW range (and higher). One example of this is Masdar's Nour One project in Abu Dhabi, which is expected to be constructed in 2012/2013

A new 150 kW installation in Southern California, setting a new size record for the use of SolarWorld's Sunkits solar system. See news on page 5 (image courtesy of SolarWorld).

- *Finsterwalde* (**Q-Cells**, 80 MW);
- *Solarpark Ernsthof* (LDK modules, relatio as developer, 34.5 MW);
- *Solarpark FinowTower* (**Suntech Power** modules, **Solarhybrid AG** as developer, 24.24 MW);
- *Solarpark Thüngen* (**Conergy** modules and developer, 18 MW).

These have to be seen beside several other large scale ground mounted projects, which have already been installed, like *Strasskirchen* (54 MW) or *Lieberose* (53 MW).

Italy's market almost reached 3.5 GW of cumulated installed PV power in 2010. This is a major increase compared to the total installed PV power at the end of 2009 (1.1 GW). During 2010 about 2,300MW was installed, an increase of 190% compared to the 795 MW installed in 2009.

However, at the moment the future outlook for Italy's PV market is difficult to quantify. The country's new *Conto Energia III* sees funding for solar electricity reduce dramatically, starting from 2011. However, solar PV expert Paul Gipe wrote in a recent article that the Italian Government's detailed new tariffs for solar PV still target 23,000 MW by 2017. The new target supersedes the previous 8,000 MW target that was likely to be surpassed this year.

And according to Gipe's article, while many commentators have emphasised that the new policy "cuts" the existing tariffs dramatically, Italian solar PV tariffs will still remain among the highest in Europe, relative to the more intense insolation in Italy. And the country's recent decision to abandon any development of nuclear power will see even more focus placed on renewable energy going forward.

France overachieved in 2010 with a total of 720 MW installed, leading to cumulatively installed PV installations of just over 1 GW. In March 2011 a new FiT was announced, which more or less prohibits facilities of more than 100 kWp when planned independently. To protect the large-scale PV market in France against this strategy, commercial sites will be tendered in accordance with the Government's goals for yearly annexes. FiTs for privately-owned small-scale systems (less than 100 kWp) will be revised on a 3-monthly basis, and adjusted according to the previous ruling.

In 2010 a total of 1.5 GW of new capacity was installed in the **Czech Republic**. However, the country has cut tariffs in half for facilities above 30 kWp. This makes the forecast for 2011 poor, with an estimated 350 MW of newly-installed capacity. This would be equal to a drop of 70%.

Following market meltdown in 2009, **Spain**'s PV market regained some momentum - with around 370 MW connected in 2010. This is an increase of almost 2000% compared to the miniscule 17 MW installed in 2009. However, even 370 MW is still only slightly above 10% of what was achieved in Spain's boom years: back in 2008 Spain was one of the most important markets for the PV industry, with more than 3.3 GW installed. The recent recovery is more due to lower component prices and favourable irradiation. But capacity growth of more than 1 GW a year could be a thing of the past.

The **U.S.** nearly doubled the amount of newly-installed capacity, and ended up installing just under 1 GW in 2010. As there is no overall U.S. market for solar PV, markets are more or less State-based, sometimes being driven in a region by individual utility companies.

2011 and beyond: just who produces the highest efficiency monocrystalline modules?

When it comes to judging the quality of PV modules, individual module 'efficiency' is often cited as a benchmark to compare the performance of different modules. With all the talk of laboratory world-records and 'quality' modules by the renowned brands, it's easy to lose sight of the real picture of module efficiency. Which module manufacturers produce the most efficient Monocrystalline PV modules currently available on the market? Solarplaza (*http://www.solarplaza.com*) has just published a comparison that looks at the relative efficiencies of different monocrystalline (mc-Si) modules (*see table below*).

As many would have expected, **Sunpower** sells the solar module with the highest efficiency, at 19.6%. The technology of number two on the list – **Sanyo** – is actually not based on mc-Si alone. Sanyo's *HIT* technology combines a crystalline silicon cell with an additional amorphous silicon layer. The difference between the leader and the number 10 on the list is 3.4%; in absolute terms a small difference, but relatively significant. The difference could mean that a consumer needs less space, for example on a roof, to achieve the same system power or energy yield. It could therefore mean less Balance of System (BOS) investments.

Whether it pays out to use the higher efficient modules is yet to be seen. The ranking does not say anything about the module cost, the relative cost per Watt peak power; or about the most important topic, the cost per produced solar kWh. It could mean that a lower efficiency module is much less expensive, so that the additional cost of other components (frame, cables, installation work etc) can easily be covered by the money saved on the modules.

Also, some modules perform better in real life conditions rather than in the standard test condition mentioned on the certified datasheet.

And what about the origin of the modules? Europeans, as well as Americans, often claim that their products offer the "best quality", but this does not show in terms of technology superiority. Although the number one, **Sunpower**, is from the U.S., its cells and modules are actually produced in the Philippines. Apart from **Crown Renewable Energy**, all the other modules are from Asia. And it's interesting to see that some of the more well known brands did not make it into the Top-10. Also interesting to see is the relatively young and unknown brands that are listed.

	Manufacturer	Module efficiency	Module type
1.	**Sunpower**	19.60%	E19 / 320 SOLAR PANEL
2.	**AUO**	19.50%	PM318B00
3.	**Sanyo Electric**	19.00%	HIT-N240SE10
4.	**Jiawei**	18.30%	JW-S100
5.	**Crown Renewable Energy**	18.30%	Summit 100LM
6.	**JA Solar**	16.84%	JAM5(L)-72-215/SI
7.	**Trina Solar**	16.40%	TSM-210DC80
8.	**CNPV Solar**	16.20%	CNPV-105M
9.	**Yingli Solar**	16.20%	Panda 265 Series
10.	**Jetion**	16.20%	JT315SAc

Top ten efficient modules: This table was first published on 27 June 2011 and was updated on 5 July 2011: Data was collated from public sources such as product datasheets online, by SolarPlaza (http://www.solarplaza.com)

With expected drops in important markets all over Europe, the hope for the U.S. is substantial growth for solar PV. The U.S. has excellent solar resource in many States, as well as availability of land for PV facilities (though there remain big problems with permitting). Colorado moved into the top five states for installed capacity in 2010, behind Arizona and Nevada. California still constitutes the major market (almost 30% of new installations) and New Jersey remained at number two with about 15%.

Markets to watch remain India and China. **India** plans to have a total of 22 GW of solar PV installed by 2022, starting from a low base of only a few MWs. This could also help to electrify rural areas within India, where 40% of households still lack basic access to electricity.

China will start investing in solar PV not only to help its burgeoning module supply industry (which has flooded the global market in recent times, leading to oversupply), but also to reduce its dependence on gas, oil and coal. China has a vast potential solar resource.

Pricing of PV projects look set to fall further

Even though PV technology costs have steadily fallen, the PV industry needs to drive down the **levelised cost of electricity (LCOE)** of the technology even further, especially in light of declining incentives.

In terms of manufacturing costs, module prices declined another 20%–30% in 2010, while the cost of other components stayed more or less the same. Since modules make up about 50%-60% of PV systems' costs, this had a substantial impact on system costs.

As of spring 2011 for example, German companies involved in EPC projects (in the low MW range) were offering prices of around €2,100 per kilowatt peak. And even lower EPC prices for larger size projects could be possible moving forward.

Of course, outside of Germany EPC prices remain higher due to differing market conditions, but take into account the expected downsize in

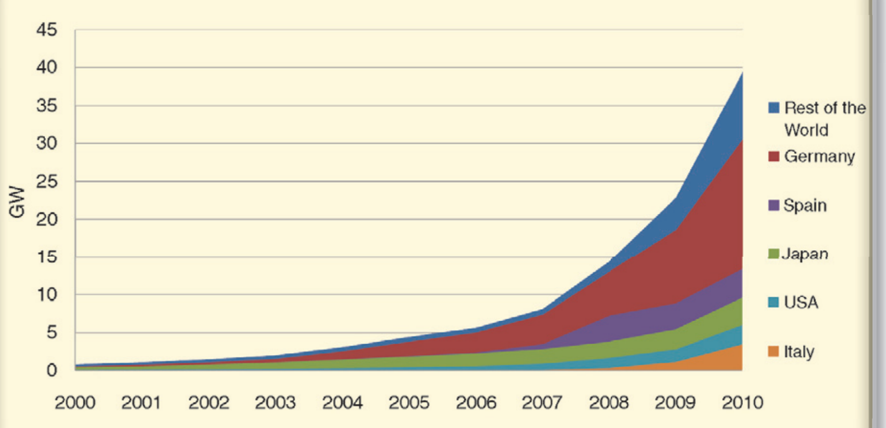

Figure 1: Historical development of global cumulative PV power installed per country (up until the end of 2010)

demand due to slowing markets in Europe, and the highly-competitive global PV market for modules, and further declines in system prices globally seem likely. Whether these system cost reductions will compensate for falling FiT revenue is uncertain, but falling system prices - while difficult for short term profitability - are ultimately a good thing for the industry.

As in previous years, crystalline silicon-based modules (mono- and polycrystalline) made up the largest share of solar PV modules manufactured, about 90% (around 10% thin film). In terms of **newly-installed production capacity added in 2010**, thin film accounted for around 20%, giving it a cumulative share of 10%-20% in the global marketplace (at the end of 2010).

Other technologies (such as Concentrating Photovoltaics - CPV - still have not achieved a noticeable market share). However, a major breakthrough for the CPV market was achieved in early 2011, when **Concentrix** (now **Soitec**) was chosen to provide CPV technology for a huge solar plant in Southern California: **Tenaska Solar Ventures** plans to build a *Solar Energy Center (ISEC WEST)*, which will provide 150 MW of power when it is completed in 2015. The use of Fresnel lenses – which concentrate direct solar irradiation on highly-efficient CPV cells – may kick start the technology from laboratory-scale research to utility-scale applications.

Competition among the module suppliers was the main driver behind the decline in module prices. Asian module suppliers priced aggressively, particularly those coming from China. In Germany for example, the majority of modules sold in 2010 were imported from Asia (China), whereas only around 12% were physically manufactured in Germany.

Thin film modules continued to push onto the global markets, with **First Solar** continuing its relentless push during 2010. Going forward, CIS and CIGS modules will soon be offered in meaningful quantities by several global players such as **Bosch Solar**, **Q-Cells** and **Solar Frontier**, having improved efficiency rates (above 12%).

However, it will be interesting to see when the inevitable decline in the cost of silicon will take away one of the crucial competitive advantages of thin film, and what effect that has on technology selection. What seems likely is that, as always, each of the technologies will continue to have its own niche markets, whether that be location or application; on grid or off grid, utility scale or rooftop.

Regarding global production capacities:

- PV supply capability (so called **nameplate capacity**) exceeded demand in 2010. Total global production capacity for modules worldwide was between 33 and 36 GW, far beyond the newly-installed capacity of around 16.6 GW in 2010;
- Global production capacity for silicon was around 350,000 tonnes;
- Global production of wafers was between 30 and 35 GW in 2010, of which more than 55% was located in China; Germany accounted for around 10% of wafer production capacity;
- Crystalline Silicon cell production is now mainly located in Asia. In 2010 it was estimated to be around 27-28 GW, of which 50% was located in China. Module production of c-Si was slightly higher and was probably in the range of 30 to 32 GW; In 2010, global thin film production reached around 3.5 GW.

(*Source - EPIA: Global Market Outlook for Photovoltaics until 2015; 2011*)

	Cumulated installed capacity 2010 (GW)		Installed capacity 2010 (GW)		Estimated electricity generation 2010 (TWh/y)
Europe		30.2		14.7	38.9
North America		2.8		1.3	5.5
South America		˜0		˜0	<0.1
Asia		5.1		1.5	11
Africa		0.1		˜0	0.1
Oceania		0.5		0.3	1.5
World total		**39.5**		**16.6**	**˜50**
Largest national Market	Germany	17.2	Germany	7.4	12.5

Table 4: Summary of the worldwide solar PV market status, up until the end of 2010 (*source: EPIA – 2015 Global Market Outlook for Photovoltaics until 2015; experts' estimates*)